SafeAlert: Theory of Change

Core Assumption, What Could Break It, How the Pilot Tests It

SafeAlert Project · May 2026 · One-page working document

Core assumption

Nigerian fintech companies are beginning to integrate general-purpose language models into customer-facing and fraud-related workflows. These models are selected and deployed without any evaluation specific to Nigerian fintech fraud patterns. **If a language model is evaluated against prompts that represent real Nigerian fintech scam patterns before it is deployed, the results will reveal safety gaps that the deploying team can address, and a model that passes the evaluation is meaningfully safer to deploy than one that has not been tested.**

This assumption rests on four sub-claims that are each individually plausible but not individually guaranteed.

#	Sub-claim	Statement	Confidence level
A1	**Prompt coverage**	A dataset of 220 prompts across 8 scam categories is sufficient to reveal meaningful safety gaps in a model's behaviour.	*Moderate. Coverage is real but limited.*
A2	**Score validity**	Refusal rate and classification accuracy, as defined by the scoring rubric, are meaningful proxies for real-world safety performance.	*Moderate. Proxy measures have known limitations.*
A3	**Generalisability**	Results from GPT-4o mini and Llama 3.1 8B at temperature 0 are informative about how those models would behave in a deployed Nigerian fintech context.	*Lower. Deployed models use system prompts and RAG not tested here.*
A4	**Actionability**	Fintech procurement teams and compliance officers can use SafeAlert results to make better model selection decisions.	*Higher. The gap in pre-deployment testing is well documented.*

What could break it

Three failure modes would invalidate the theory of change, partially or entirely. Each is stated honestly here so the pilot report can address them directly.

> **Failure mode 1: The synthetic dataset does not reflect real scam patterns**
> If the 220 prompts do not represent how Nigerian fintech scams actually appear in the wild, then a model passing the evaluation may still fail when deployed against real messages, and a model failing the evaluation may actually perform well in practice. The dataset is grounded in documented fraud categories and reviewed by Andrew against real fraud case summaries, but it remains synthetic. A model optimised against this dataset would not necessarily be safe in the real world.

> **Failure mode 2: Base model behaviour does not predict deployed model behaviour**
> SafeAlert tests models at the API level without a system prompt (pre-remediation) and with a generic safety system prompt (post-remediation). A deployed fintech chatbot uses a product-specific system

prompt, a retrieval-augmented knowledge base, and possibly fine-tuning. The base model's refusal behaviour may differ substantially from the deployed model's behaviour in either direction: a model that passes the SafeAlert test may still produce harmful content in a specific deployment configuration, and a model that fails may be made safe through careful system prompt design. This is the most significant structural limitation of the project.

Failure mode 3: Procurement teams do not use the results

Even if the kit produces valid and informative results, the theory of change only holds if a fintech fraud team or compliance officer actually runs it before making a procurement decision. Adoption requires that the kit is accessible, the implementation guide is clear enough to follow without ML expertise, and the results are interpretable without specialised knowledge. If the kit is too difficult to run or the outputs are not actionable, the evaluation findings do not translate into safer deployments.

How the pilot tests it

The pilot provides evidence about the theory of change in the following ways. It does not prove the theory is correct, but it makes it more or less plausible depending on the results.

What the pilot tests	How it tests it	What the result tells us
Whether the dataset reveals real gaps (A1, A2)	Run 220 prompts against GPT-4o mini and Llama 3.1 8B pre-remediation. If both models score perfectly on all prompts, the dataset is not discriminating enough. If results vary across categories and models, the dataset is revealing genuine differences in model behaviour.	Results that vary meaningfully across scam categories and models support A1 and A2. A ceiling effect (all prompts passed) would undermine both.
Whether a simple remediation step improves results (A2, A3)	Compare pre and post-remediation metrics. A measurable improvement in refusal rate and TPR after adding the system prompt would suggest that the evaluation is sensitive to meaningful interventions.	A measurable delta supports A2 and provides the first evidence for A3. No delta would suggest the evaluation is not sensitive enough or that the system prompt is insufficient.
Whether the kit is runnable without ML expertise (A4)	The implementation guide is tested implicitly by the pilot itself. If Hadiza can run the full notebook and Andrew can follow the scoring rubric without external support, the kit meets the accessibility requirement for A4.	A smooth pilot run supports A4. Blockers, ambiguities in the rubric, or notebook failures that require debugging provide specific improvements for v2.
Whether failure modes 1 and 2 are bounded	The pilot report documents all false negatives with concrete examples. If false negatives cluster around specific prompt structures or scam categories, this provides evidence about where the dataset needs strengthening (failure mode 1) and where base model behaviour is least predictive (failure mode 2).	Clustered failure patterns are more actionable than random ones and give a clearer picture of how far the results generalise to real deployments.

Honest statement of scope

> The pilot cannot prove that SafeAlert makes deployments safer. It can show that the kit discriminates meaningfully between models, that a simple intervention improves measurable safety indicators, and that the kit is runnable by a non-specialist team. Those three things together make the theory of change plausible. Demonstrating real-world deployment impact would require a follow-on study with a fintech partner, which is outside the scope of this competition submission.

This document is referenced in the pilot report under Limitations and Theory of Change (Section 5).